

Course Number	COE 328
Course Title	Digital Systems
Semester/Year	Semester 3, Year 2
Section Number	1

Lab 2 - Function implementation and minimization

Assignment Title	Lab 1
Submission Date	September 28, 2022
Due Date	October 2nd, 2022

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(Note: Remove the first 4 digits from your student ID)

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In-lab:

Assigned Function:

This lab's results and simulation are based on function 13 as the assigned function. The chosen student number is listed as follows: 501103255, where the last 4 digits were used. In this case, the last 4 digits are 3-2-5-5. Therefore, the assigned function is:

$$f = \sum(3, 2, 5, 5, 11, 12, 15)$$

Truth table:

Since we are given the minterms listed in the above function, a truth table can be made, where the given variables are listed as “1” in the output.

	<u>INPUT</u>				<u>OUTPUT</u>
MINTERMS	A	B	C	D	F
m0	0	0	0	0	0
m1	0	0	0	1	0
m2	0	0	1	0	1
m3	0	0	1	1	1
m4	0	1	0	0	0
m5	0	1	0	1	1
m6	0	1	1	0	0
m7	0	1	1	1	0
m8	1	0	0	0	0
m9	1	0	0	1	0
m10	1	0	1	0	0
m11	1	0	1	1	1
m12	1	1	0	0	1
m13	1	1	0	1	0
m14	1	1	1	0	0
m15	1	1	1	1	1

Student # : 501103255

Function : 13

$$F_{13} = \sum (3, 2, 5, 5, 11, 12, 15)$$

K-map

AB \ CD	00	01	11	10
00	0	0	1	1
01	0	1	0	0
11	1	0	1	0
10	0	0	1	0

Simplified Logic Equation:

$$f = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D} + \bar{A}B\bar{C}\bar{D} + A\bar{B}C\bar{D} + A\bar{B}\bar{C}\bar{D} + A\bar{B}C\bar{D}$$

$$= \bar{A}\bar{B}C(\bar{D} + D) + \bar{A}B\bar{C}\bar{D} + A\bar{B}\bar{C}\bar{D} + A\bar{B}C\bar{D}$$

$$= \bar{A}\bar{B}C + \bar{A}B\bar{C}\bar{D} + A\bar{B}\bar{C}\bar{D} + A\bar{B}C\bar{D}$$

Figure 1: Screenshot of VHDL Code:

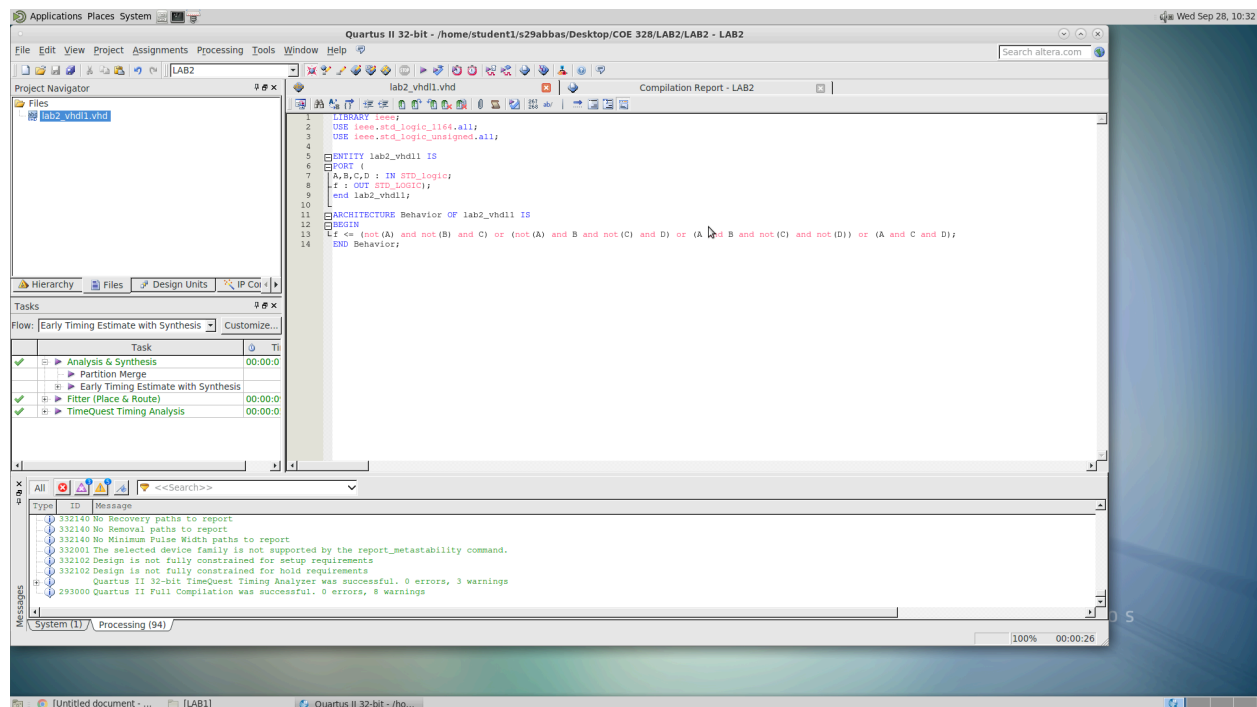


Figure 2: Screenshot of successful compilation

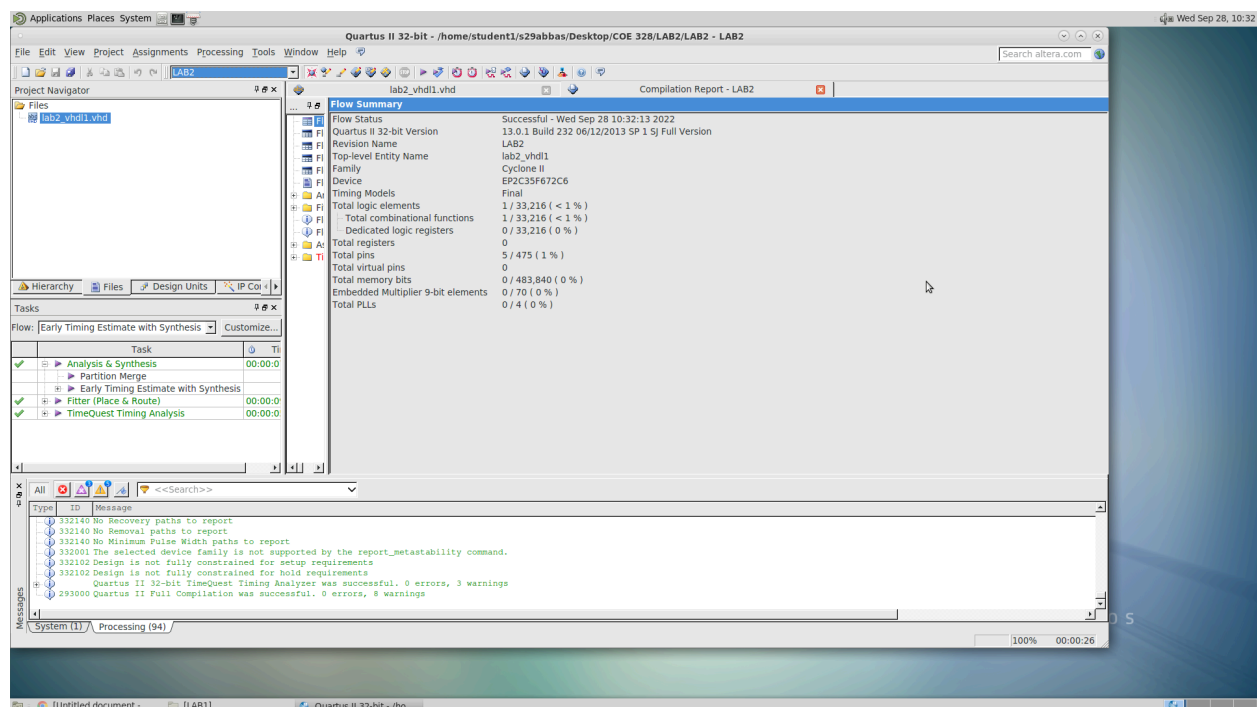
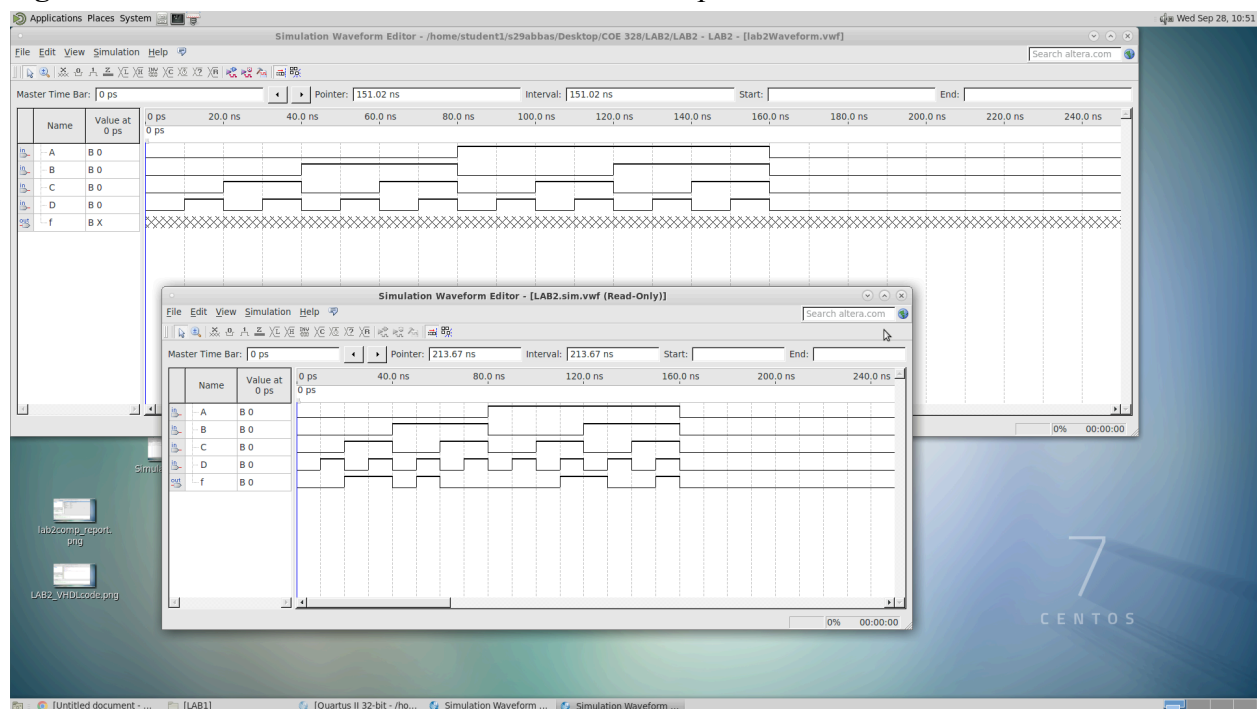


Figure 3: Screenshot of waveform simulation and output



Conclusion:

In conclusion, this lab describes the process of interpreting a function using a truth table, expressing the sum of products (SOPs) given the minterms, simplifying a logic equation to construct a Karnaugh map (K-map), and simulating a waveform through a VHDL code. Understanding and implementing a function is an essential skill to further enhance the efficiency and consistency of any given circuit. Therefore, this lab proves to be beneficial for future implementations of this topic.